FOREX Rate Prediction using Chaos and Quantile Regression Random Forest

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***Abstract*— This paper presents a hybrid of chaos modeling and Quantile Regression Random Forest (QRRF) for Foreign Exchange (FOREX) Rate prediction. The exchange rates data of US Dollar (USD) versus Japanese Yen (JPY), British Pound (GBP), and Euro (EUR) are used to test the efficacy of proposed model. Based on the experiments conducted, we conclude that the proposed model yielded accurate predictions compared to Chaos + Quantile Regression (QR), Chaos+Random Forest (RF) and that of Pradeepkumar and Ravi [12] in terms of both Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE).**

Keywords*— *FOREX Rate Prediction; Hybrid model; chaos; QR; RF; QRRF;

I. INTRODUCTION

The FOREX market is the major and the most profitable financial market [1]. The FOREX Rate is an important economic index in the international monetary market. Its optimal prediction helps a country to revisit its economic policies and make necessary changes so that the trade relationships are properly maintained there by making its economy stronger. So, FOREX Rate prediction can never be underestimated [2].

The financial time series including exchange rate are intrinsically noisy, non-stationary and deterministically chaotic [3]. A chaotic time series is a time series with the characteristics such as non-linearity, determinism and sensitivity to initial conditions [4]. The chaotic time series prediction engages with the prediction of future behavior of chaotic system by utilizing the current and past states of that system. Chaos theory always demands us to expect the unexpected. Therefore, prediction of FOREX Rate is always a complex problem.

It is popular that a financial time series can be forecast after it is embedded in a state space and later, it is combined with supervised learning mechanism [5,6]. Chaos theory helps the forecaster in this context by embedding scalar time series in a state space using optimal parameters such as lag and embedding dimension, where lag is the time delay and embedding dimension is the minimum number of variables needed to represent the dynamics of the chaotic system [7,8].

The Quantile Regression provides a framework to model the relationship between a dependent variable and its covariates using conditional quantile functions. The QRRF is a variant of random forest designed to optimize the prediction of quantiles . In the past, Ravi and Sharma [9] proposed a hybrid of support vectors and QRRF to solve regression problems. QRRF allows us to predict a continuous variable without the need to know the form of the underlying distribution. Therefore, we propose a novel Chaos+ QRRF hybrid to forecast FOREX rates and compare its performance with that of Chaos+QR and Chaos+RF hybrids.

The following study forms background to the current work. Pradeepkumar and Ravi [12] predicted FOREX Rate using two two-stage hybrid models. These models need two stages of modeling in order to yield accurate predictions. However, the current work yields the accurate predictions with just one stage of modeling. However, in the proposed model, as the preprocessing step, the FOREX Rate time series is embedded in a state space with the help of optimal embedding parameters as chaos is present in the data. Later, a QR-based hybrid, QRRF, accepts the newly modeled data and yields the accurate predictions.

The remainder of this paper is organized as follows: The Section II presents Chaos-based hybrid forecasting models. The overview of Chaos+QRRF is presented in the Section III. The Section IV describes experimental methodology. The Section V discusses the results obtained. The concluding remarks are presented in Section VI.

II. RELATED WORK

In literature there are very few chaos-based hybrid models for FOREX Rate prediction. In 2003, Pavlidis et al. [10] presented a hybrid methodology for prediction of financial time series. This methodology works through five stages. (i) The optimal embedding dimension for constructing phase space is calculated using False Nearest Neighbors (FNN) method. (ii) The time delayed vectors are clustered using unsupervised K-windows algorithm. (iii) A different feedforward ANN is trained using the DE algorithm and PSO algorithm on each cluster separately. (iv) The test set patterns

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identified are allotted to the clusters. (v) For each test pattern, predictions are generated using the trained networks. The proposed methodology yielded promising results for both DE and PSO than single feed-forward neural network (FNN) after experimenting with daily exchange rates of USD-JPY and USD-GBP in terms of mean predictive accuracy. It also can be investigated for expanded test set also. It also needs to be tested for multi-step prediction and for other economic and chaotic time series.

Then, Huang et al [11] implemented a Chaos-SVR model for FOREX rate prediction. During first stage, the base time series are converted to phase space using both optimal lag and embedding dimension. In the second stage, SVMs are constructed for forecasting. The authors claimed that the Chaos-SVR model extracted the important features of FOREX rate dynamics and performed better after experimenting with daily exchange rates of USD-EUR, GBP, NZD, AUD, JPY and RUB in terms of RMSE, MSE and MAE. It can also be tested with other financial data. Its results can be used to invest in global market carefully.

Finally, Pradeepkumar and Ravi [12] proposed two intelligent two-stage hybrids of FOREX Rate prediction models: ANN (MLP/GRNN/GMDH) + PSO/ Polynomial Regression (PR) and PSO+ANN/PR. In the proposed models, the Stage-1 obtains initial predictions, while Stage-2 fine tunes them. Both these models checked for the presence of chaos at two stages and phase space in both stages is reconstructed using both optimal lag and embedding dimension. The modeling of chaos in the data helped the proposed models in yielding better predictions compared to the stand-alone forecasting models after experimenting with daily exchange rates of USD-JPY, USD-GBP and USD-EUR in terms of both MSE and MAPE.

Quantile regression found very few applications in financial time series prediction. Taylor [13] proposed quantile regression approach to estimate the conditional probability distribution of multi-period financial returns from a range of different holding periods. The proposed approach is compared to GARCH-based quantile estimates using daily exchange rates and concluded that the proposed method is best suitable for estimating the conditional density. Similarly, Huang et al. [14] adopted the QR to forecast volatility from nine exchange rates using 19 years of data and concluded that QR produced more reliable volatility forecasts.

Apart from this, to the best of our knowledge, there are no other works reported in literature that predicted FOREX rates using quantile regression-based hybrids. Further, there is also no other forecasting model that forecast FOREX Rates using a combination of chaos theory and quantile regression hybrids. Our work bridges that gap in literature

III. Proposed Model

A. Overview of techniques used

In the proposed hybrid model, the Saida's method is used to determine whether chaos is present in the data or not. The Akaike Information Criterion (AIC) is used for determining optimal lag. The Cao's method is used to obtain minimum embedding dimension. The forecasting models, viz., QR, RF and QRRF are used to obtain predictions.

i) Saida's Method

Saida [15] proposed a method to check the presence of chaos in a time series using Lypunov exponent λ as in (1):

$$\lambda = \lim_{t\to\infty} \frac{1}{t} \ln \frac{|f(A_0,t)|}{|A_0|} \tag{1}$$

Where $f(A_0,t)$ is a function of the location of initial values, t is time and A_0 is initial value.

This method returns λ, which is either 1 or 0. If return value is 1, chaos is absent; else, it is present.

ii) *Cao's Method*

Cao [16] proposed a method for obtaining minimum embedding dimension for a given set. Let $A_1, A_2, ..., A_N$ be a time series. It can be reconstructed as time delay vectors in the phase space as in (2):

$$A_m(i) = [A(i), A(i+t), A(i+2t), ..., A(i+m-1)t], \quad i = 1,2,..., n-(m-1)t \tag{2}$$

Where m is the embedding dimension and t is the time delay. Here $A_m(i)$ is the ith reconstructed vector and $A_m^{NN}(i)$ is the nearest neighbor of D_m in embedding dimension m as in (3):

$$A_m^{NN}(i) = [A^{NN}(i), A^{NN}(i+t), A^{NN}(i+2t), ..., A^{NN}(i+(m-1)t) \tag{3}$$

Similar to the idea of false nearest neighborhood (FNN) method, they define (4)

$$g_2(i,m) = \frac{\left\| A_{m+1}(i) - A_{m+1}^{NN}(i) \right\|}{\left\| A_m(i) - A_m^{NN}(i) \right\|} \tag{4}$$

Here ||.|| is the Euclidian distance and is given by the maximum norm here. $A_{m+1}(i)$ is the ith reconstructed vector and $A_{m+1}^{NN}(i)$ is its nearest neighbor in embedding dimension $m+1$. To avoid the problems of FNN, they define the mean value of all $g_2(i,m)$ s as in (5):

$$E(m) = \frac{1}{n-ml} \sum_{i}^{n-ml} g_2(i,m) \tag{5}$$

$E(m)$ is only dependent on the dimension m and lag l. to investigate its variation from m to $m+1$, we define

$$E_1(m) = \frac{E(m+1)}{E(m)} \tag{6}$$

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It is found that $E_1(m)$ stops changing when m>m_0, the attractor is reconstructed in m_0-dimensional phase space and m_0 is the minimum embedding dimension is what we needed.

iii) Akaike Information Criterion (AIC)

Akaike Information Criterion [17] can be used to obtain optimal lag *l* from time series data. In Vector Auto Regression (VAR), lagged values of the predictor variable are to be included. More lag values are needed when the far past values of predictor variable are still affecting today's value. AIC can be used to determine the lag or time delay. It chooses lag length *l* to minimize $\frac{\log(SSR(l))}{N}+(l+1)\frac{c(N)}{N}$ where *SSR(l)* is the sum of squared residuals for the VAR with lag length *l*, n is the number of observations and c(N) = 2.

iv) Quantile Regression

Quantile (**τ**) is marked as one boundary of equally sized consecutive subsets. For example, the median (50th percentile), 70th percentile, 90th percentile…etc are different quantiles. Quantile Regression [18] is an approach that can be used to examine the behavior of the response variable at different quantiles other than its average Gaussian distribution. This examination results in different conclusions compared to examining only the average of response variable.

v) Random Forest

In order to understand random forest, we first need to appreciate how Classification and Regression tree (CART) [19] works. To solve classification and regression problems, CART, a non-parametric and statistical technique, is very much useful [19]. It can be constructed as a binary decision tree by recursively partitioning the data. It is able to deal with non-linearity and its results are easily interpretable. However, it is not stable though there are small changes in training data. A bootstrap aggregating technique namely bagging can be applied to overcome this problem [20]. In bagging, a single tree is fit to each bootstrap sample from a large number of bootstrap samples taken from the dataset. By using each of the fitted trees, the predictions for new data are received and their prediction trees are averaged. It turns out that RF [21] is a modified version of bagged tree. In RF, for each tree and at each node, a random subset of the predictors is used which is not possible with bagging. This procedure results in a robust model. It also handles the missing data effectively.

vi) Quantile Regression Random Forest

Quantile Regression Random Forest (QRRF), introduced by Meinshausen [22], is a generalization of RF. In general, each node in an RF keeps only the mean of the observations that fall into a node. Each node does not consider other information other than mean. In contrast, QRRFs can keep the value of all observations in a node. In other words, they consider the spread of the predictor variable. It allows the construction of various prediction intervals which cover highly probable new observations.

B. Chaos+QRRF

In this hybrid model, first the phase space is reconstructed using chaos parameters such as lag and embedding dimension. Then the reconstructed data is input to QRRF in order to obtain predictions.

Let $A=\{A_1,A_2,...,A_k,A_{k+1},...,A_N\}$ be a time series data of N observations at times 1, 2, …, N respectively. Then the proposed Chaos+QRRF proceeds as follows.

Step 1: Check A for the presence of chaos and Rebuild it using both lag *l* and embedding dimension *m*, if chaos is present.
Step 2: Divide A as training set with time $t=lm+1,lm+2,...,k$ and test set with $t=k+1,k+2,...,n$ respectively.
Step 3: Input the chaotic modeled training set and Quantile values $Q^*=\{Q_1, Q_2 \dots Q_m\}$
Step 4: For each quantile value **τ** in Q^*, repeat steps (5), (6) and (7):
Step 5: Train QRRF using $f(A_t)$ where $f(.)$ is a non-linear function that maps output variable A_t with input variables A_{t-l}, A_{t-2l}, A_{t-3l} … A_{t-ml} and obtain predictions of training set P_t with $t=lm+1,lm+2,...,k$.
Step 6: Input chaotic modeled test set to trained QRRF.
Step 7: obtain test set predictions P_t where t=$k+1, k+2,..., N$.

IV. Experimental Design

In the current work, the FOREX Rate data obtained from (http://www.federalreserve.gov/releases/h10/hist/) is used. The daily US dollar exchange rates with respect to three currencies-JPY, GBP and EUR are collected. The daily data of both USD-JPY and USD-GBP from 1st January 1993 to 31st December 2013 and USD-EUR from 3rd January 2000 to 31st December 2013, are used as datasets. For each of dataset, 80% of dataset is used as training set and 20% of dataset is used as test set as given in Table. I. It also presents various descriptive statistical measures of the datasets such as mean, median, standard deviation, maximum, minimum, skewness and kurtosis.

The performance measures used are Mean Squared Error (MSE) and Mean Absolute Percentage Error (MAPE) as defined in (7) and (8):

$$MSE=\frac{\sum_{t=1}^{N}(A_t-P_t)^2}{N} \tag{7}$$

$$MAPE=\frac{100}{N}\sum_{t=1}^{N}\left|\frac{A_t-P_t}{A_t}\right| \tag{8}$$

where N is count of predicted observations, A_t is actual observation at time *t*, P_t is predicted value obtained at time *t*. The MSE measures the average of the square of the error and the MAPE measures the size of the error in percentage terms.

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TABLE I. DESCRIPTIVE STATISTICS FOR ALL DATASETS

Data	No. of observations	Mean	Median	Min	Max	SD	Skewness	Kurtosis
USD-JPY								
All data	6036	110.55	110.27	75.72	159.90	16.93	0.03	-0.17
Training set	4829	116.30	116.00	81.12	159.90	13.28	0.46	0.54
Test set	1207	87.52	86.20	75.72	105.20	7.99	0.27	-1.28
USD-GBP								
All data	6036	1.655	1.615	1.366	2.110	0.16	0.75	-0.3
Training set	4829	1.674	1.635	1.366	2.110	0.17	0.48	-0.79
Test set	1207	1.579	1.586	1.397	1.698	0.05	-0.63	0.23
USD-EUR								
All data	3772	1.219	1.269	0.827	1.601	0.19	-0.44	-0.74
Training set	3018	1.189	1.220	0.827	1.601	0.2	-0.12	-0.97
Test set	754	1.336	1.326	1.206	1.488	0.06	0.38	-0.3

TABLE II. RESULTS OF HYBRID MODELS FOR USD-JPYDATA

Model	Training Set MSE (MAPE)	Test set MSE (MAPE)
Pradeepkumar and Ravi [9]	0.7314 (0.5487)	0.3548 (0.4757)
Chaos+QR (l=4, m=20, τ = 0.5)	0.6577 (0.5033)	0.3490 (0.4690)
Chaos+RF (l=4, m=20)	0.6983 (0.5253)	20.5370 (4.1299)
Chaos+QRRF (l=4, m=20, τ = 0.5)	**0.2029 (0.1067)**	**0.1695 (0.4608)**

TABLE III. RESULTS OF HYBRID MODELS FOR USD-GBP DATA

Model	Training Set MSE (MAPE)	Test set MSE (MAPE)
Pradeepkumar and Ravi [9]	0.0001097 (0.4595)	0.00094 (0.4773)
Chaos+QR (l=5, m=16, τ = 0.5)	0.00010211 (0.4312)	0.00083193 (0.4442)
Chaos+RF (l=5, m=16)	0.00010875 (0.4489)	0.00012917 (0.5464)
Chaos+QRRF (l=5, m=16, τ = 0.5)	**3.6484e-09 (0.0011)**	**5.9879e-08 (0.0067)**

TABLE IV. RESULTS OF HYBRID MODELS FOR USD-EUR DATA

Model	Training Set MSE (MAPE)	Test set MSE (MAPE)
Pradeepkumar and Ravi [9]	0.0000684 (0.5147)	0.0000736 (0.4870)
Chaos+QR (l=1, m=10, τ = 0.5)	0.00006235 (0.4857)	0.000063817 (0.4418)
Chaos+RF (l=1, m=10)	0.00007663 (0.5515)	0.000076894 (0.5028)
Chaos+QRRF (l=1, m=10, τ = 0.5)	**2.5565e-09 (0.0015)**	**2.0875e-08 (0.0056)**

V. RESULTS AND DISCUSSION

Various tools are used to obtain results. The Saida's method implemented in MATLAB® (http://www.mathworks.in/matlabcentral/fileexchange/22667-chaos-test) is used to check the presence of chaos. The optimal lag is obtained using Gretl® tool (http://gretl.sourceforge.net/) and the Cao's method implemented in MATLAB® (http://www.mathworks.in/matlabcentral/fileexchange/36935-minimum-embedding-dimension/content/cao_deneme.m) to obtain minimum embedding dimension. Various packages in R such as 'quantreg', 'randomForest' and 'quantregForest' are used to implement chaos-based QR, RF and QRRF respectively.

In Tables II, III and IV, the MSE and MAPE values for both training set and test set of datasets USD-JPY, USD-GBP and USD-EUR respectively are presented along with the optimal lag (l), the optimal embedding dimension (m) and the median quantile (τ = 0.5) considered. From these tables, it is clear that Chaos+QRRF outperformed all other Chaos-based forecasting models, viz., Chaos+QR, Chaos+RF, and Pradeepkumar &

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Ravi in terms of both MSE and MAPE. . It can be easily observed that the Chaos+QRRF yielded the best predictions compared to other forecasting models specified above because of the following reasons:

- While QR models the heteroskedasticity in the data, RF is very robust because of its inherent strength in terms of random subspace sampling and bagging. Further, QRRFs are consistent for conditional quantile estimation and can choose appropriate number of random covariates to find the best split, which, in turn helps it to yield better predictions than other forecasting models.
- The bagging procedure in QRRF can reduce the variance in the predictions, which, in turn, led to better predictions.
- QRRFs consider random feature subsets of data at different quantiles, construct trees for each subset over a bootstrap sample and ensemble the obtained results. Therefore, QRRF yielded better predictions than other forecasting models.

Figures 1-3 depict the comparisons of various hybrid models using predictions of test sets of all datasets The forecasting accuracy measures, MSE and MAPE, however, do not formally test whether progresses in forecast accuracy between two forecasting models are statistically significant or not. To test the statistical significance, a popular test proposed by Diebold and Mariano [23] is used here. The test is implemented as part of 'forecast' package in R. The results of the DM Test are presented in Table V. If the test statistic is less than or equal to 0.05, then the corresponding model is statistically not significantly different from Chaos+QRRF. Accordingly, it is clear that Chaos+QRRF is statistically superior to all models in all cases.

TABLE V. RESULTS OF DM TEST OF ALL DATASETS

Model *Chaos+QRRF vs.*	**USD-JPY**	**USD-GBP**	**USD-EUR**
Pradeepkumar and Ravi [11]	7.027803	1.506212	0.7128105
Chaos+QR	6.986417	0.830984	0.6375361
Chaos+RF	25.01626	1.29057	0.7682258

VI. CONCLUSION

The paper proposes Chaos-based quantile regression random forest viz., Chaos+QRRF to predict FOREX Rates. The results obtained on test datasets indicate that the proposed hybrid model outperformed the methods of Pradeepkumar & Ravi [11], Chaos+QR and Chaos+RF in terms of both MSE and MAPE. As chaos in the datasets modeled using optimal embedding parameters systematically, the application of QRRF on rebuilt dataset yielded best predictions. As this hybrid yields best predictions, its utility can also be explored in other financial time series.

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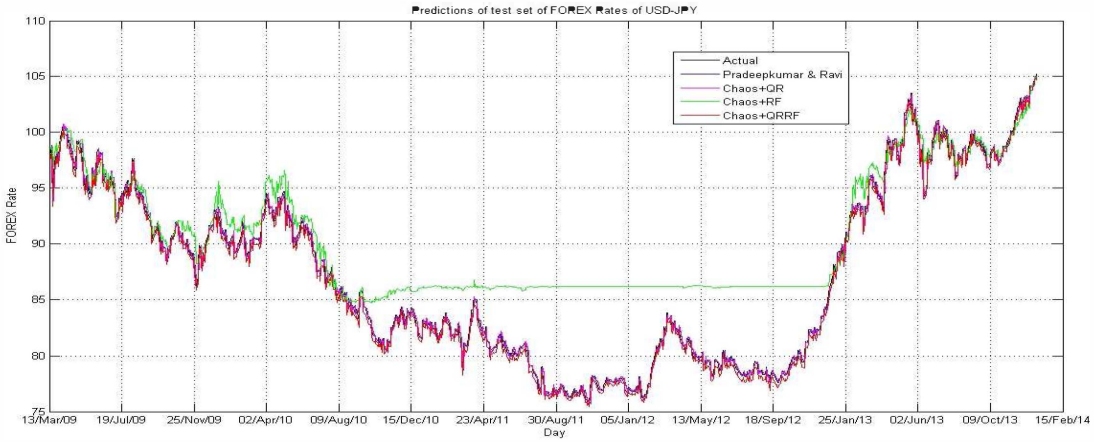


Fig. 1. The comparison of different models with predictions of Test set of FOREX Rates of USD-JPY

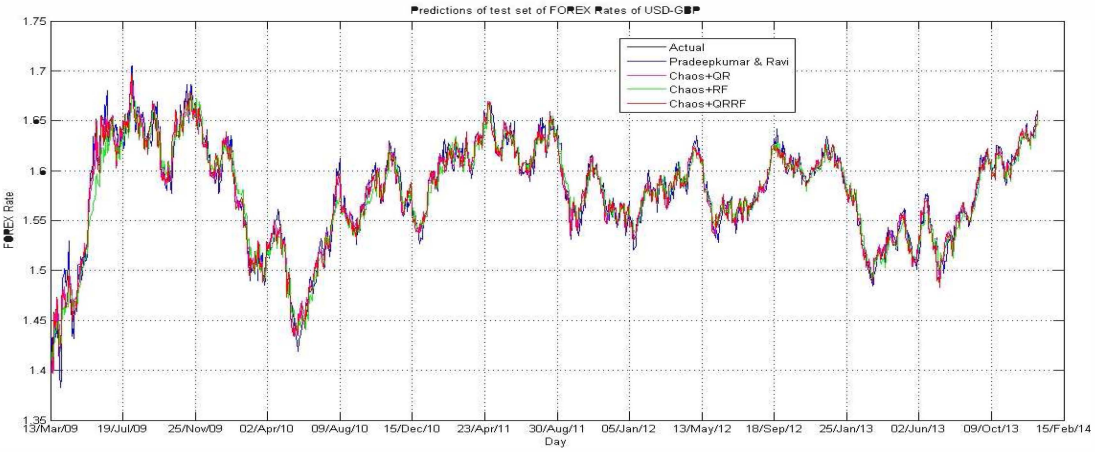


Fig. 2. The comparison of different models with predictions of Test set of FOREX Rates of USD-GBP

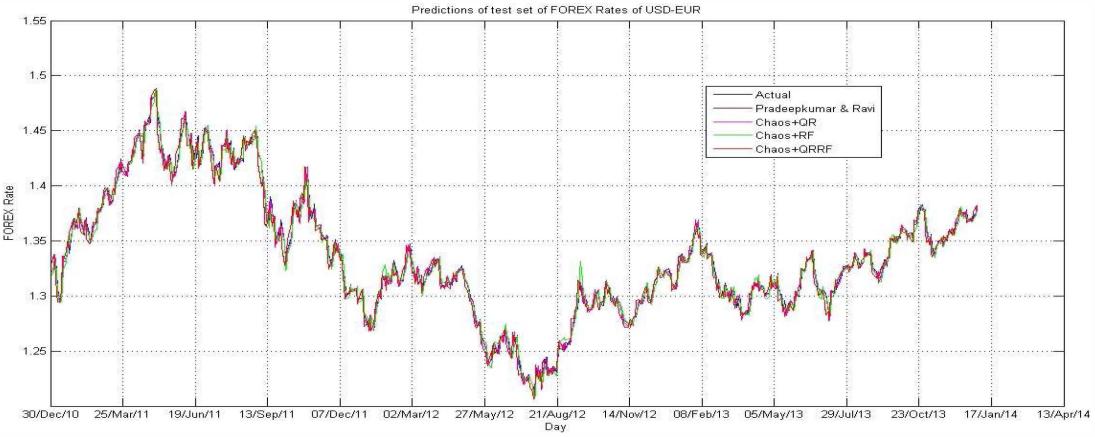


Fig. 3. comparison of different models with predictions of Test set of FOREX Rates of USD-EUR

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